yeast001 — shot list

Every shot in the assembly, in order. **118 shots, 13:03.09**, 23.976p, 1920×1080.

Generated from `picks.json`, so it cannot drift from the timeline. *Sound* is where the audio comes from; *picture* is only filled in where the shot is a split edit and the picture comes from somewhere else.

Sources

Clip	Shots	On screen
`MVI_0040.MOV`	49	5:28.04
`MVI_0038.MOV`	23	2:36.03
`MVI_0045.MOV`	18	2:00.05
`MVI_0037.MOV`	11	1:06.10
`MVI_0041.MOV`	10	1:01.16
`MVI_0042.MOV`	4	0:42.05
`MVI_0043.MOV`	2	0:05.03
`MVI_0036.MOV`	1	0:03.11

Music

Track	In	Out	Level
Audiio_AsTallAsPine_ATAP_TwinLakes	0:00.00	0:29.23	16 dB under the voice
Audiio_SK_Commercial_TwoBigReasons30	12:31.06	13:03.09	16 dB under the voice

Shots

#	At	Dur	Sound	Picture	What it is
1	0:00.00	2.2s	`MVI_0043.MOV @ 13.43`	**Best Yeast Broll.MOV @ 4.00**	hook VO — yeast bubbling in action
2	0:02.05	2.9s	`MVI_0043.MOV @ 19.52`	**Katie Dancing.MOV @ 56.81**	hook VO — way too excited about yeast
3	0:05.03	5.3s	`MVI_0040.MOV @ 668.79`	—	hook claim — you don't need sugar, you need time

#	At	Dur	Sound	Picture	What it is
4	0:10.10	3.5s	`MVI_0036.MOV @ 35.20`	—	name + promise (take 4)
5	0:13.21	13.2s	`MVI_0037.MOV @ 19.02`	—	yeast is really old — Egypt, Asia, Europe
6	0:27.02	3.0s	`MVI_0037.MOV @ 32.91`	—	it's around, it grows on grapes
7	0:30.03	7.4s	`MVI_0038.MOV @ 211.79`	—	yeast in the air, on the grapes — not inoculated
8	0:37.12	2.3s	`MVI_0038.MOV @ 219.64`	—	yeast in the air, on the grapes — not inoculated (2)
9	0:39.18	4.8s	`MVI_0038.MOV @ 222.35`	—	yeast in the air, on the grapes — not inoculated (3)
10	0:44.12	6.5s	`MVI_0038.MOV @ 228.19`	—	yeast in the air, on the grapes — not inoculated (4)
11	0:51.00	1.1s	`MVI_0038.MOV @ 235.61`	—	yeast in the air, on the grapes — not inoculated (5)
12	0:52.02	1.8s	`MVI_0037.MOV @ 66.02`	—	brewers → friends making doughs
13	0:53.21	6.2s	`MVI_0037.MOV @ 68.40`	—	brewers → friends making doughs (2)
14	1:00.02	11.1s	`MVI_0037.MOV @ 75.16`	—	brewers → friends making doughs (3)
15	1:11.05	1.6s	`MVI_0037.MOV @ 99.47`	—	bakers get bubbly batches — passed back and forth
16	1:12.20	7.0s	`MVI_0037.MOV @ 101.73`	—	bakers get bubbly batches — passed back and forth (2)
17	1:19.20	3.0s	`MVI_0038.MOV @ 254.50`	—	selective breeding (the term)
18	1:22.21	6.7s	`MVI_0037.MOV @ 213.00`	—	they may not know what they're breeding — dud batch
19	1:29.13	3.8s	`MVI_0037.MOV @ 224.81`	—	successful batch passed on — genes selected
20	1:33.07	4.6s	`MVI_0037.MOV @ 229.10`	—	successful batch passed on — genes selected (2)

#	At	Dur	Sound	Picture	What it is
21	1:37.22	7.4s	`MVI_0037.MOV @ 234.23`	—	successful batch passed on — genes selected (3)
22	1:45.07	0.6s	`MVI_0038.MOV @ 311.02`	—	until Louis Pasteur — gooseneck, microbes
23	1:45.21	10.8s	`MVI_0038.MOV @ 312.10`	—	until Louis Pasteur — gooseneck, microbes (2)
24	1:56.15	3.3s	`MVI_0038.MOV @ 325.37`	—	until Louis Pasteur — gooseneck, microbes (3)
25	1:59.22	7.7s	`MVI_0038.MOV @ 329.54`	—	until Louis Pasteur — gooseneck, microbes (4)
26	2:07.15	8.6s	`MVI_0041.MOV @ 30.49`	—	aside — glass bulb, stretches a neck
27	2:16.05	12.6s	`MVI_0041.MOV @ 41.12`	—	open here, gooseneck shape — heats it to kill what's inside
28	2:28.20	2.7s	`MVI_0041.MOV @ 104.73`	—	if spontaneous generation were true, the food breaks down
29	2:31.12	6.4s	`MVI_0041.MOV @ 108.02`	—	if spontaneous generation were true, the food breaks down (2)
30	2:37.22	5.4s	`MVI_0041.MOV @ 115.16`	—	germ theory — germs get stuck in the downspout
31	2:43.08	7.4s	`MVI_0041.MOV @ 121.29`	—	germ theory — germs get stuck in the downspout (2)
32	2:50.18	1.4s	`MVI_0041.MOV @ 129.17`	—	germ theory — germs get stuck in the downspout (3)
33	2:52.04	4.9s	`MVI_0041.MOV @ 134.84`	—	nothing happens to the food — and that's what happens
34	2:57.02	3.8s	`MVI_0041.MOV @ 146.81`	—	Pasteur proves there is no spontaneous generation
35	3:00.22	8.4s	`MVI_0041.MOV @ 151.32`	—	Pasteur proves there is no spontaneous generation (2)
36	3:09.07	16.5s	`MVI_0038.MOV @ 414.62`	—	we meet Saccharomyces cerevisiae
37	3:25.18	9.2s	`MVI_0038.MOV @ 431.60`	—	we meet Saccharomyces cerevisiae (2)

#	At	Dur	Sound	Picture	What it is
38	3:34.22	6.9s	`MVI_0038.MOV @ 441.57`	—	a fungus, single cell, thrives in fertile environments
39	3:41.20	4.0s	`MVI_0038.MOV @ 449.16`	—	a fungus, single cell, thrives in fertile environments (2)
40	3:45.19	11.3s	`MVI_0042.MOV @ 1.42`	—	aside — sourdough uses yeast too, plus lactic bacteria
41	3:57.01	8.1s	`MVI_0042.MOV @ 13.30`	—	aside — sourdough uses yeast too, plus lactic bacteria (2)
42	4:05.03	8.4s	`MVI_0042.MOV @ 25.53`	—	setting sourdough aside — S. cerevisiae is repeatable
43	4:13.12	14.5s	`MVI_0042.MOV @ 34.37`	—	setting sourdough aside — S. cerevisiae is repeatable (2)
44	4:28.00	1.2s	`MVI_0038.MOV @ 548.97`	—	from a mother ferment to measured yeast on hand
45	4:29.05	11.4s	`MVI_0038.MOV @ 552.34`	—	from a mother ferment to measured yeast on hand (2)
46	4:40.15	6.0s	`MVI_0038.MOV @ 564.27`	—	from a mother ferment to measured yeast on hand (3)
47	4:46.14	2.0s	`MVI_0038.MOV @ 570.95`	—	from a mother ferment to measured yeast on hand (4)
48	4:48.14	6.7s	`MVI_0038.MOV @ 573.82`	—	from a mother ferment to measured yeast on hand (5)
49	4:55.06	4.1s	`MVI_0038.MOV @ 581.29`	—	our way of baking and brewing changes
50	4:59.09	9.7s	`MVI_0038.MOV @ 595.60`	—	because we can be intentional — measure the organism
51	5:09.02	16.0s	`MVI_0038.MOV @ 649.65`	**MVI_0026.MOV @ 3.00**	used all over — big brewers to at-home enthusiasts
52	5:25.02	7.1s	`MVI_0045.MOV @ 22.31`	—	off camera I have examples of yeast you might find
53	5:32.05	7.2s	`MVI_0045.MOV @ 35.12`	**MVI_0027.MOV @ 6.01**	active dry, pre-seeded with sugar — you don't need it
54	5:39.09	2.0s	`MVI_0045.MOV @ 59.23`	**MVI_0028.MOV @ 1.50**	instant yeast right here

#	At	Dur	Sound	Picture	What it is
55	5:41.09	9.6s	`MVI_0045.MOV @ 62.73`	**MVI_0028.MOV @ 3.42**	instant — smaller, wakes faster
56	5:50.23	3.5s	`MVI_0045.MOV @ 72.82`	**MVI_0031.MOV @ 1.00**	fresh yeast — fragrant, straight into a dough, short shelf life
57	5:54.12	3.2s	`MVI_0045.MOV @ 77.16`	**MVI_0031.MOV @ 4.55**	fresh yeast — fragrant, straight into a dough, short shelf life (2)
58	5:57.17	8.7s	`MVI_0045.MOV @ 81.41`	**MVI_0031.MOV @ 7.76**	fresh yeast — fragrant, straight into a dough, short shelf life (3)
59	6:06.09	9.6s	`MVI_0045.MOV @ 107.86`	—	eats through its food and goes dormant — dried yeast lasts
60	6:15.23	4.1s	`MVI_0045.MOV @ 118.24`	—	eats through its food and goes dormant — dried yeast lasts (2)
61	6:20.01	3.8s	`MVI_0045.MOV @ 122.79`	—	eats through its food and goes dormant — dried yeast lasts (3)
62	6:23.21	6.6s	`MVI_0045.MOV @ 127.13`	—	eats through its food and goes dormant — dried yeast lasts (4)
63	6:30.12	6.2s	`MVI_0045.MOV @ 134.13`	**MVI_0029.MOV @ 2.00**	mead yeast packet — it's just S. cerevisiae, all the same bug
64	6:36.16	3.1s	`MVI_0045.MOV @ 141.10`	**MVI_0029.MOV @ 8.17**	mead yeast packet — it's just S. cerevisiae, all the same bug (2)
65	6:39.18	0.9s	`MVI_0045.MOV @ 145.06`	**MVI_0029.MOV @ 11.26**	mead yeast packet — it's just S. cerevisiae, all the same bug (3)
66	6:40.16	6.3s	`MVI_0045.MOV @ 146.48`	—	mead yeast packet — it's just S. cerevisiae, all the same bug (4)
67	6:47.00	8.0s	`MVI_0045.MOV @ 154.20`	—	mead yeast packet — it's just S. cerevisiae, all the same bug (5)
68	6:54.23	19.0s	`MVI_0045.MOV @ 168.46`	—	a different way to reach the brewer — same thing in the baking aisle
69	7:13.23	11.3s	`MVI_0045.MOV @ 187.60`	—	someone with a marketing degree — it's all the same S. cerevisiae
70	7:25.07	4.3s	`MVI_0040.MOV @ 31.61`	—	why so many names, so many formats?
71	7:29.14	1.4s	`MVI_0040.MOV @ 45.30`	—	the answer is access and need

#	At	Dur	Sound	Picture	What it is
72	7:30.23	32.2s	`MVI_0040.MOV @ 46.88`	—	industrial bakers vs the farm in Kansas
73	8:03.03	1.0s	`MVI_0040.MOV @ 79.58`	—	industrial bakers vs the farm in Kansas (2)
74	8:04.03	4.2s	`MVI_0040.MOV @ 89.76`	—	fresh, dry — active dry and instant
75	8:08.07	10.3s	`MVI_0040.MOV @ 122.79`	—	why doesn't the large baker use instant?
76	8:18.14	10.4s	`MVI_0040.MOV @ 136.09`	—	processing damages the yeast — shows up in elasticity
77	8:28.23	9.3s	`MVI_0040.MOV @ 148.06`	—	of the gluten — shears it, dries the bread faster
78	8:38.07	3.7s	`MVI_0040.MOV @ 172.38`	—	glutathione (the term)
79	8:42.00	3.9s	`MVI_0040.MOV @ 183.22`	—	gets into the gluten network — shorter texture, sandwich bread
80	8:45.22	0.8s	`MVI_0040.MOV @ 187.60`	—	gets into the gluten network — shorter texture, sandwich bread (2)
81	8:46.16	6.2s	`MVI_0040.MOV @ 189.02`	—	gets into the gluten network — shorter texture, sandwich bread (3)
82	8:52.20	0.9s	`MVI_0040.MOV @ 195.65`	—	gets into the gluten network — shorter texture, sandwich bread (4)
83	8:53.18	5.0s	`MVI_0040.MOV @ 197.36`	—	gets into the gluten network — shorter texture, sandwich bread (5)
84	8:58.17	1.3s	`MVI_0040.MOV @ 208.25`	—	fresh yeast is consistent — at home you might not notice
85	9:00.00	6.3s	`MVI_0040.MOV @ 210.00`	—	fresh yeast is consistent — at home you might not notice (2)
86	9:06.08	0.6s	`MVI_0040.MOV @ 217.18`	—	fresh yeast is consistent — at home you might not notice (3)
87	9:06.23	7.6s	`MVI_0040.MOV @ 218.26`	—	fresh yeast is consistent — at home you might not notice (4)
88	9:14.13	3.5s	`MVI_0040.MOV @ 248.79`	—	it's about what's available

#	At	Dur	Sound	Picture	What it is
89	9:18.00	11.8s	`MVI_0040.MOV @ 254.50`	—	not worth driving two hours for a day longer
90	9:29.20	7.3s	`MVI_0040.MOV @ 276.53`	—	how does it reproduce?
91	9:37.03	6.5s	`MVI_0040.MOV @ 328.24`	—	mother cell buds a daughter — mitosis, two cells
92	9:43.16	5.4s	`MVI_0040.MOV @ 335.25`	—	mother cell buds a daughter — mitosis, two cells (2)
93	9:49.01	5.5s	`MVI_0040.MOV @ 373.54`	—	grows where there are nutrients — sugars and starches
94	9:54.14	5.9s	`MVI_0040.MOV @ 379.71`	—	grows where there are nutrients — sugars and starches (2)
95	10:00.12	12.5s	`MVI_0040.MOV @ 387.47`	—	amylases break starch into sugars
96	10:12.23	8.0s	`MVI_0040.MOV @ 409.03`	—	sucrose with invertase
97	10:21.00	1.7s	`MVI_0040.MOV @ 444.32`	—	simple sugars, zymase → CO_2 and alcohol
98	10:22.16	1.3s	`MVI_0040.MOV @ 446.61`	—	simple sugars, zymase → CO_2 and alcohol (2)
99	10:24.00	4.8s	`MVI_0040.MOV @ 448.53`	—	simple sugars, zymase → CO_2 and alcohol (3)
100	10:28.19	6.8s	`MVI_0040.MOV @ 466.38`	—	the byproducts we want — alcohol, or the rise we bake off
101	10:35.13	6.8s	`MVI_0040.MOV @ 473.60`	—	the byproducts we want — alcohol, or the rise we bake off (2)
102	10:42.09	12.3s	`MVI_0040.MOV @ 492.24`	—	the right environment — between 50 degrees dormant and 140 killed off
103	10:54.15	8.2s	`MVI_0040.MOV @ 505.09`	—	the right environment — between 50 degrees dormant and 140 killed off (2)
104	11:02.20	5.1s	`MVI_0040.MOV @ 518.10`	—	the sweet spot — 93 to 100-and-a-couple — where we proof
105	11:07.22	16.4s	`MVI_0040.MOV @ 523.94`	—	the sweet spot — 93 to 100-and-a-couple — where we proof (2)

#	At	Dur	Sound	Picture	What it is
106	11:24.07	6.8s	`MVI_0040.MOV @ 551.01`	—	salt, fat, spices get in the way — slows proving
107	11:31.02	11.0s	`MVI_0040.MOV @ 558.85`	—	salt, fat, spices get in the way — slows proving (2)
108	11:42.02	3.6s	`MVI_0040.MOV @ 583.12`	—	sugar speeds it up
109	11:45.16	7.8s	`MVI_0040.MOV @ 625.12`	—	active dry says add sugar — you don't have to, let it sit longer
110	11:53.11	4.8s	`MVI_0040.MOV @ 633.38`	—	active dry says add sugar — you don't have to, let it sit longer (2)
111	11:58.05	15.1s	`MVI_0040.MOV @ 648.19`	**MVI_0023.MOV @ 14.01**	yeast just wants flour and water — the pre-ferment, bubbling
112	12:13.06	3.8s	`MVI_0040.MOV @ 674.88`	—	adding sugar speeds it up, but then — sweeter dough
113	12:17.01	7.3s	`MVI_0040.MOV @ 682.10`	—	if that's not what you want, you don't need to add sugar
114	12:24.08	3.7s	`MVI_0040.MOV @ 700.03`	**MVI_0025.MOV @ 20.02**	a pre-ferment is a version of a sourdough — easier to add later
115	12:28.01	11.3s	`MVI_0040.MOV @ 704.50`	**MVI_0025.MOV @ 23.73**	a pre-ferment is a version of a sourdough — easier to add later (2)
116	12:39.09	4.0s	`MVI_0040.MOV @ 716.01`	—	it also, as the yeast breaks down the flour,
117	12:43.10	4.8s	`MVI_0040.MOV @ 722.26`	—	creates a more complex flavor
118	12:48.04	15.2s	`MVI_0038.MOV @ 709.83`	—	close — the same little guy, factory cakes to store packets

What each shot says

1. 0:00.00 — This is yeast bubbling in action.

2. 0:02.05 — And this is me way too excited about yeast.

3. 0:05.03 — You don't actually need sugar to make your yeast grow. You just need time.

4. 0:10.10 — Hi, I'm Katie. I'm a food scientist, and this is my deep dive into yeast.

5. 0:13.21 — Yeast is really old. We've had it around for at least a few thousand years. We've found it in archaeological records in Egypt. We have found it in archaeological records in parts of Asia and in Europe.

6. 0:27.02 — It's around. It's in our environment. It grows on grapes.

7. 0:30.03 — We know that the yeast is in the air around all these ancient people, and so it is just falling into their food, or it's on the grapes that they're smashing up to make wine. It is being pulled in from the environment. It's not being inoculated. It's not being pushed into the food by the bakers themselves until they figure out kind of what's going on, that you need a little bit of something to make something else.

12. 0:52.02 — You get the brewers that are taking grapes and grains and turning it into wine and beer and mead. And they're producing these fermenting bubbly batches of goodness that they like, and they are taking it and they're passing it on to their friends that are making doughs.

15. 1:11.05 — those bakers are getting bubbly batches that are making dough that rises and are different than flat breads and it's being passed back and forth back and forth.

17. 1:19.20 — With the passing back and forth, you get selective breeding going on.

18. 1:22.21 — They may not know what they're breeding, but they're being selective about it. If it's a dud batch, they're not passing it on to their friend.

19. 1:29.13 — If they have a successful batch, they're passing it on to their friend, and the genes of the yeast are being selected for the ones that ferment and make alcohol and raise bread and do lots of things that these ancient people wanted.

22. 1:45.07 — It wasn't until we meet Louis Pasteur that we have an inkling of what's making one successful batch work and what's being passed on between these bakers and brewers. He has a gooseneck experiment and disproves spontaneous generation and says there are microbes, little bugs that we can't see all around the world.

26. 2:07.15 — If you don't know about the Louis Pasteur experiment, what he does is he takes a glass bulb and stretches out a neck on it.

27. 2:16.05 — that is open over here, but it has a gooseneck shape. And he heats it up to kill off any bacteria that's on the inside with some food that he has cooked, that he has put inside of this gooseneck bottle.

28. 2:28.20 — And so if there's any truth to spontaneous generation, the food will just break down. There will be no problem with it doing what food has always done.

30. 2:37.22 — But if the food breaks down because of germ, germ theory, what he was thinking of is germs can't get in through the gooseneck bottle. It falls down. It gets stuck in the downspout right here.

33. 2:52.04 — so nothing will happen to the food. It won't break down. It won't do anything. And that's what happens.

34. 2:57.02 — And so Louis Pasteur proves that there is no spontaneous generation, that there is something in the air, bacteria, yeast, living organisms that are getting into our food and making changes.

36. 3:09.07 — meet our Saccharomyces cerevisiae. And this is the most common form of yeast that we use today. There are other yeasts that are used in sourdoughs, but for the most part, when you get to industrial

bakers and you get to at-home bakers and you get to brewers, they're all using the same little bug, Saccharomyces cerevisiae.

38. 3:34.22 — Saccharomyces cerevisiae is a yeast which is a member of the fungi family. It's a single cell organism that lives and thrives and grows in fertile environments.

40. 3:45.19 — Now, an aside about sourdough. Sourdough is also using yeast. Most sourdoughs are using Saccharomyces cerevisiae, but you'll find in other sourdoughs that there are lots of other different types of yeast involved. It's also using a bacteria that produces lactic acid.

42. 4:05.03 — And so I'm setting sourdough to the side so we could specifically talk about yeast, but sourdough is using yeast. It is using other forms of yeast that could do the same thing as Saccharomyces cerevisiae, but Saccharomyces cerevisiae is specifically known for producing repeatable results that do what we want it to do, which is why we use it so much.

44. 4:28.00 — And with the discovery of Saccharomyces cerevisiae, we move from always needing to have a sourdough-based model where we have a... mother dough, mother ferment, maintaining our yeast that we pull from to make our doughs, to being able to have on hand just the yeast and using that and introducing it into our dough at measured amounts that we want.

49. 4:55.06 — And our way of baking and our way of brewing changes

50. 4:59.09 — because we can be a little bit more intentional with how we are baking and how we are brewing with being able to measure out the organism that we're introducing into our batches.

51. 5:09.02 — Saccharomyces cerevisiae is used all over the industries from alcohol industries to our baking industries, from the big brewers, the big wine makers, the big bakers to the at-home enthusiasts.

52. 5:25.02 — here off camera I have a couple examples of yeast that you might find when you're out and about looking to buy yeast.

53. 5:32.05 — I have some active dry yeast that they actually pre-seeded with sugar. You don't actually need the sugar in there,

54. 5:39.09 — I have instant yeast right here.

55. 5:41.09 — It's a little bit smaller and so it takes a little bit less for the water to get in and get the yeast going and active and ready to be added into a dough.

56. 5:50.23 — I also have a fresh yeast right here. It's ready to go. It's very fragrant. It can just be added straight into a dough. It does have a shorter shelf life.

59. 6:06.09 — It's more likely to eat through the food source that it has built in with it when it is sold. And so it doesn't last as long on the shelf as say the active dry and the instant, which the producers of the active dry and instant have dried it out and made the yeast cells dormant and less likely to die off.

63. 6:30.12 — i even have this yeast packet right here that is marketed as for mead and when you look at it and it's in ingredients it is just Saccharomyces cerevisiae all of these things that i just showed you are the same all of these are the same little bug the same microbe the same yeast fungi they're all Saccharomyces cerevisiae

68. 6:54.23 — it's a different way of trying to get it into the hands of those who are looking for it because when you're a brewer you might be looking for brewers yeast because you know that's what i'm doing

that's what i'm looking for but if you went to the store and bought baking yeast it would be the same thing but it might not click that that is what you're looking for.

69. 7:13.23 — And so someone with a marketing degree, with a business degree is saying, hey, let me sell to you. I know what you need. It's this. But it's really all just the same Saccharomyces cerevisiae.

70. 7:25.07 — Why do we have so many different names for it? Why do we have so many different formats for it?

71. 7:29.14 — The answer is access and need.

72. 7:30.23 — You have industrial bakers who don't necessarily want to take the time to reactivate something from being dormant that want to just use it and move on and get going with their timelines. and you have the at-home baker in the middle of a farm in Kansas that needs something every once in a while and doesn't want to have to drive two hours back and forth to a grocery store every couple days to get fresh yeast. They need something on hand that they can keep that won't die off on them as frequently.

74. 8:04.03 — So you have fresh yeast and dry yeast and you have active dry and instant.

75. 8:08.07 — why doesn't the large baker use the instant yeast it doesn't need to be reactivated it doesn't take as long to be reactivated and it's always on hand

76. 8:18.14 — But in the processing of the active dry yeast and the instant yeast, the yeast gets a little damaged and it can show up in the elasticity

77. 8:28.23 — of the gluten development. It can shear it a little bit and it might help dry, it helps dry out the bread a little bit faster.

78. 8:38.07 — because of this chemical release called glutathione

79. 8:42.00 — glutathione will get into the gluten network and have it break a little bit easier and it will help the starch release the water a little faster so you have a more of a shorter texture that you might get more of in a sandwich bread and have it dry out faster

84. 8:58.17 — so having a fresh yeast allows these bakers to have a product that does what they want it to do. It's consistent. It doesn't dry out. And to be honest, an at-home baker might not care. You might not notice. They consume the bread within a couple days because they're only making one loaf at a time.

88. 9:14.13 — so it's about what's available and what you can use and what you can find

89. 9:18.00 — and what other things are going on in your life for what matters because it might not be worth driving a couple hours to a grocery store to get something to have a bread that's a little bit stretchier that lasts a day longer.

90. 9:29.20 — if we have so many of the same little Saccharomyces cerevisiae, how do we reproduce it? What's going on?

91. 9:37.03 — So you have a mother cell that grows and buds on itself a daughter cell. And when that daughter cell is big and identical to the mother cell, it uses mitosis and splits into two cells.

93. 9:49.01 — yeast will continue to grow and reproduce itself when it's in an environment that has lots of nutrients for it. And those nutrients are sugars and starches. It breaks down starch into sugar.

95. 10:00.12 — So starch itself will break, flour itself uses an alpha amylase and a beta amylase to break those starches down into sugars that the yeast can digest.

96. 10:12.23 — the yeast will start digesting sucrose with invertase and break that down and use that energy for itself

97. 10:21.00 — the yeast will use simple sugars and break it down with the enzyme zymase into CO2 and alcohols.

100. 10:28.19 — and the CO2 and the alcohol are the byproducts that we're looking for as humans, either in alcohol production or in helping our bread rise with the CO2, giving the gas that we will eventually bake off.

102. 10:42.09 — yeast is looking for a right environment for it to thrive and to grow and that environment has lots of sugars and starches and is in a temperature between 50 degrees where it kind of shuts down and goes dormant and 140 degrees when it's completely killed off

104. 11:02.20 — There's a nice sweet spot right in the middle, about 93 degrees Fahrenheit and 100 plus a couple degrees Fahrenheit. And in that sweet temperature spot, it will grow and thrive. And that is the temperature we generally want our bread to be sitting at to kind of reproduce and proof at the rate that we want it.

106. 11:24.07 — so when we add salt and fat and other spices to our bread doughs, what we're doing is we're putting items in the way of the yeast and its food and so it's slowing down the process in which the yeast can digest the sugars that it wants for growing and that will then slow our proving time.

108. 11:42.02 — the availability of sugar will speed up the proofing time.

109. 11:45.16 — this is why in active dry yeast they recommend that you add a little bit of sugar to kind of start the yeast up and going and getting it bubbling you don't have to actually do that you can just let it sit a little longer

111. 11:58.05 — yeast itself just really wants flour and water and you can do that sometimes by getting a pre-ferment. This is what I have going right here and it's already starting to bubble up and it's just flour and water and yeast and the yeast is in a happy spot.

112. 12:13.06 — Adding sugar does speed it up, but then you end up with a sweeter dough,

113. 12:17.01 — and if that is not the final product that you want, you do not actually need to add sugar to your dough to get the yeast to grow.

114. 12:24.08 — the other byproduct of starting with maybe a pre-ferment, a pre-ferment is basically a different version of a sourdough. It lets the yeast get going and multiply and kind of creates an environment that's easier to add to your dough later.

116. 12:39.09 — It also, as the yeast breaks down the flour,

117. 12:43.10 — it creates a more complex flavor that might be what you want in your end product.

118. 12:48.04 — Saccharomyces cerevisiae is the same little guy that is being used all over the food industry from the big yeast cakes that you have in the factories to the little packets that we buy at the store for at-home baking.